Causal canonical modeling applied to real-world data

October 12, 2026

1 Overview

The goal of this paper is to apply a causal generative model, a *causal canonical model*, that can approximate observational and interventional distributions in any unknown causal model with continuous treatment and outcome variables, to real-world data (adapted to simulate causal effects). The model itself comes from the work of Zhang and Bareinboim. Specifically, the data to be used comes from the *International Stroke Trial database (version 2)*.

2 Motivation

The paper *Causal Canonical Modeling For Confounding Robust Treatment Evaluation* by Zhang and Bareinboim proposes a new method that allows researchers to approximate the observational and interventional distributions in any causal model with continuous treatment and outcome with arbitrary accuracy, using a mixture of Gaussian processes. Specifically, The model can devise a posterior distribution over parameters of the target causal effects.

In order to test the results of the paper with real-world data, a semi-synthetic model is created.

3 Dataset

The dataset that was used is *International Stroke Trial database (version 2)*, created in 1996, which aimed to establish whether early administration of aspirin, heparin, both or neither influenced the clinical course of acute ischaemic stroke. Many variables are given in the dataset, although only a few are used for the experiment.

The way that the trial was conducted was that patients who were brought into the hospital due to stroke, were given one of the following randomized treatments: aspirin, heparin, both, neither. Vitals were taken prior to treatment administration, and follow-ups were conducted 14 days and 60 days from the start of the treatment.

4 Procedure

With code provided by Dr. Zhang that implements the findings of the original paper, the only thing to do was to change the input from the example model to an adapted model, using a well-known medical database, with simulated causal effects.

5 Adapted model and simulated causal effects

In order to devise an adapted model with a continuous treatment and outcome, the following adapted treatment X and outcome Y are defined:

1. $X = f(\mathbf{Z}) + \varepsilon_x = c_1\text{Age} + c_2\text{RDELAY} + c_3\text{RSBP} + c_4\text{SEX} + \varepsilon_x = C^\top \mathbf{Z} + \varepsilon_x$
2. $Y = f(X) + \varepsilon_y = \alpha tanh(X) + b_1\text{Age} + b_2\text{RDELAY} + b_3\text{RSBP} + b_4\text{SEX} + \varepsilon_y = \alpha tanh(X) + B^\top \mathbf{Z} + \varepsilon_y$

Where $\varepsilon_x \sim N(0, \sigma_x^2)$, $\varepsilon_y \sim N(0, \sigma_y^2)$, and $Z \sim N(\mathbf{0}, \mathbf{I})$.

X, the treatment, is defined as a function of the set of covariates $\mathbf{Z}$, plus some zero-mean gaussian noise. It can be interpreted as a "treatment intensity" score.

Y is defined as a function of X, plus some zero-mean gaussian noise. It can be interpreted as a "recovery score", influenced by treatment intensity and patient characteristics. A larger value of Y indicates a better recovery, while a low value indicates a worse recovery, or even death if low enough.

The reason why we choose $f(X) = \alpha tanh(X)$ is so that is in line with this interpretation, such that higher treatment intensity provide a higher receovery score, and vice versa. This also bounds the outcome in the sense that very high values of the treatment intensity do not also cause unboundedly high recovery scores, which could introduce bias towards certain treatments.

The set of covariates $\mathbf{Z}$ are taken from the dataset and are as follows:

(Age, Delay, RSBP, Sex)

1. AGE - Patient age in years
2. RDELAY - Delay between stroke and randomisation iN hours
3. RSBP - Systolic blood pressure at randomisation (mmHg)
4. SEX - M = male; F = female

While these covariates do not function as unobserved confounders, they do introduce confounding bias as they have a causal effect on both X and Y, while X has a causal effect on Y as well.

The adapted causal graph consists of the following edges:

1. $\mathbf{Z} \to Y$
2. $\mathbf{Z} \to X$
3. $X \to Y$

Note that this causal graph does not represent the actual reality from the dataset, it is made to represent an assumed reality, using real-world data. The graph is simple enough to simulate confounding bias.

Specifically, this model is a semi-synthetic model, which functions as a hybrid between the real-world data and a simulated world, in order to effectively evaluate the result of the model with the "engineered" causal effects and ground truth that are devised from the real-world data. This assumed causal model also allows us to choose which of the variables would be the most helpful for attempting to identify causal effects in the presence of confounding bias.

Because we defined the causal graph, we know what $E[Y|X = x]$ as well as $E[Y|do(X = x)]$ are supposed to be:

To solve for $E[Y|do(X = x)]$, we force $X = x$ to be a constant, and proceed as normal.

$$E[Y|do(X = x)] = \alpha tanh(x) + B^\top E[\mathbf{Z}] + E[\varepsilon_y] = \alpha tanh(x)$$

To solve for $E[Y|X = x]$:

$$\begin{aligned} E[Y|X = x] &= \alpha E[tanh(X)|X = x] + B^\top E[\mathbf{Z}|X = x] + E[\varepsilon_y|X = x] \\ &= \alpha tanh(x) + B^\top E[\mathbf{Z}|X = x] \end{aligned}$$

Solve for $E[\mathbf{Z}|X = x]$:

We use the known fact that the conditional distribution of two multivariate normal random variables, $[Z \quad X]^\top$, is also multivariate normal, with mean:

$$\mu_{\mathbf{Z}|X=x} = E[\mathbf{Z}] + Cov(Z, X)Var(X)^{-1}(X - E[X])$$

And:

- $E[\mathbf{Z}] = 0$
- $Cov(Z, X) = \Sigma_Z C = \mathbf{I}C = C$
- $Var(X)^{-1} = \frac{1}{\|C\|_2^2 + \sigma_x^2}$
- $E[X] = C^\top E[\mathbf{Z}] + E[\varepsilon_x] = 0$

$$E[\mathbf{Z}|X = x] = \frac{C}{\|C\|_2^2 + \sigma_x^2} x$$

Plugging back into $E[Y|X = x]$, we finally get:

$$E[Y|X = x] = \alpha tanh(x) + B^\top (\frac{C}{\|C\|_2^2 + \sigma_x^2} x)$$

Note that the difference between the expected value from the interventional and observational distribution comes from the fact that X carries information about Z, so in the observational distribution, knowing X automatically gives you information about Z, which in turn influences Y as well, which is an example of confounding bias in this causal model.

6 Simulating causal effects / Data preparation

In order to simulate causal effects, the dataset was filtered to only include the covariates that are part of the adapted model (described in the previous section). Then, NA values were removed from the remaining columns of the dataset, and two new variables, X and Y, were defined.

The resulting action-reward pairs (X, Y) were then stored in the file `obs.mat`. `exp.mat` is supposed to contain values of x that are meant to represent interventional samples. In order to create this file, different samples of X are generated within the domain of $X \in \mathbb{R}$. The file helps when evaluating the learned interventional effect $E_x[Y]$.

7 Results

The goal of the original paper was to infer $E_x[Y] = E[Y|do(X = x)]$ for all possible actions x (output) from the observed samples drawn from the distribution $P(X, Y)$ (input). The hybrid model considers the case where X is not a vector, but a continuous scalar.

8 References

1. Sandercock, Peter, et al. “International Stroke Trial Database (Version 2).” Ed.ac.uk, 2025, datashare.ed.ac.uk/handle/10283/124, https://hdl.handle.net/10283/128. Accessed 24 Dec. 2025.

2. Zhang, Junzhe, and Elias Bareinboim. Causal Generative Modeling for Confounding Robust Treatment Evaluation. https://causalai.net/r131.pdf